Importing Data in R: Exploring Various Packages and Their Advantages: Data Analysis for Decision-Making (SS 2026)

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Abstract

This document describes the importance of importing data in R, introducing the packages and their advantages.

1 Why Do We Import Data?

2 Why Do We Import Data? Why with R?

- **Open-Source Accessibility**
- **The Reproducibility**
- **Auditable**
- **Efficiency and Automation**
- **Handling Complexity**

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3 Different Data Types

1. Rectangular/Tabular Data (.csv, .tsv, .txt)
2. Proprietary Spreadsheets (.xls, .xlsx)
3. Statistical Software Files (.sav, .sas7bdat, .dta)
4. Relational Databases (.db, .sqlite, .mdb)
5. Spatial/GIS Data (.shp, .tif, .kml)
6. Base Text and Fixed-Width Files (.txt, .dat, .fwf)
7. Scientific/Sensors Data (.jdx, .abs, .cnv, .nc)
8. Cloud Sheets (Google Sheet)
9. Hierarchical Data (.json)

4 Graphical and Programmatic Implementation by Data Type

4.1 Standard Tabular Data (.csv, .tsv)

- **Graphical:** In RStudio, click on File → Import Dataset, and then you can see two options for importing text file, as shown in the handout.

- **Code**:

From text Base

```
df_base <- read.csv("data.csv")
df_base
```

From text readr

```
library(readr)
df_readr <- read_csv("data.csv")
df_readr
```

4.2 Proprietary Excel Files (.xls, .xlsx)

Note: first Install the tidyverse package, then:

- **Graphical:** Select “From Excel” in the RStudio wizard. Then you can select from which sheet you want to import data, and also define a cell range (e.g., “A1:E20”) graphically.

- **Code**

```
library(readxl)
df_excel <- read_excel("file.xlsx")
df_excel
```

4.3 Statistical Software Files (.sav, .sas7bdat, .dta)

- **Graphical:** Select “From SPSS/SAS/Stata” in RStudio.

- **Code:**

```r
library(haven)
# Import SPSS file
df_spss <- read_sav("survey_data.sav")
# Import Stata file
df_stata <- read_dta("economic_data.dta")
# Preview imported data
head(df_spss)
head(df_stata)
```

4.4 Spatial / GIS Data (.shp, .tif, .kml)

- **Code:**

```r
library(sf)
# Read shapefile
spatial_data <- st_read("sample_shapefile.shp")
# Preview spatial object
print(spatial_data)
```

4.5 Base Text and Fixed-Width Files (.txt, .dat, .fwf)

- **Graphical:** Use the "From Text (base)" option in the RStudio wizard.
- **Code:**

```r
# Import fixed-width file
df_fwf <- read.fwf(
  "legacy_data.fwf",
  widths = c(3, 6, 5),
  col.names = c("ID", "Name", "Score")
)

df_fwf
```

4.6 Scientific Spectral Data (.jdx, .abs, .cnv, .nc)

- **Graphical:** Utilize specialized package GUIs or high-level functions that automate folder-wide imports Gruson et al. (2019).
- **Code:**

```r
library(lightr)

# Import spectral data
spec_data <- lr_get_spec("spectrum.jdx")

# Show spectral data
head(spec_data)
```

4.7 Cloud Sheets (Google Sheets/ URLs)

- **Code:**

```r
library(googlesheets4)
# Import Google Sheet
gs4_deauth()
sheet_data <- read_sheet(
  "https://docs.google.com/spreadsheets/d/1BoD9bXwbxm1NJWhEnk6BVXUxuxWbWArQWVT3LIKk2PQ/
)

sheet_data
```

4.8 Hierarchical Data (.json)

- **Graphical:** While RStudio does not have a dedicated "JSON Wizard" in the same way it does for Excel or CSV, users can often preview JSON structures through the File Pane or by using specialized viewing packages.
- **Code:**

```r
library(jsonlite)
df_json <- fromJSON("data.json")
df_json
```

5 The Comparison (Advantages/Limitations)

- **Tabular Data (readr):** Very fast import; creates tibbles / Limited by RAM.
- **Excel Sheets (readxl):** No Java required; supports multiple sheets / Limited scalability.
- **Statistical Files (haven):** Preserves labels and metadata / Limited newest-version support.
- **Databases -SQL(DBI):** Handles large datasets / Requires SQL knowledge.
- **Spatial / GIS (sf):** Supports CRS metadata / Complex for beginners.
- **Hierarchical (jsonlite):** Standard for APIs and JSON / Nested data can be complex.
- **Batch Import (purrr or lightr):** Automates multi-file import / Requires functional programming.
- **Cloud Services (googlesheets4):** Direct cloud access / Requires OAuth authentication.

6 Conclusion

Mastering data import is about matching the right package to the right data structure while respecting hardware limits. Choosing programmatic scripts ensures that the decision-making process is transparent, scalable, and reproducible Bivand et al. (2008), Huber (2025).

7 Class Assignment

Download the files from here link.

7.1 Help The HR Manager!

1. Your HR manager is importing personnel database every month and it is a real pain for her to create the report every time from it! She asks you to prepare a dynamic report by importing the data into R, so she can easily prepare reports more efficiently. Go ahead and do first step which is importing the exe1.csv file into R Studio.

7.2 Everyday Sales Report!

2. You are a manager in a company and the sales department gives you sales reports in excel file every morning. consider that you have received exe2.xlsx file today. Import it to Rstudio.

7.3 Spare Parts Sheet

3. Imagine that you and your colleagues in the company are working on a Google spreadsheet (Link), together. And now you want to import the data into R. Import it in the R Studio. Note: If you had problem with the link to Google spreadsheet, upload this file exe3.xlsx yourself and then use your own link. Note that you should use the Google sheet link, not Google Drive link.

8 References

Bivand, R. S., Pebesma, E. J., & Gómez-Rubio, V. (2008). *Applied spatial data analysis with R*. Springer Science & Business Media. https://doi.org/10.1007/978-0-387-78171-6

Gruson, H., White, T., & Maia, R. (2019). Lightr: Import spectral data and metadata in R. *Journal of Open Source Software*, *4*(43), 1857. https://doi.org/10.21105/joss.01857

Huber, S. (2025). *How to use R for data science*. https://hubchev.github.io/ds/